SHLOK KUMAR GOENKA

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PROFESSIONAL SUMMARY

Data Science undergraduate with a strong foundation in machine learning, deep learning, NLP, and computer vision. Proven ability to architect full-stack AI solutions across digital forensics, privacy-preserving analytics, civic intelligence, and sustainable cloud computing. Two patents cleared through institutional review (Indian IPR verification ongoing) with four additional patents in progress spanning quantum-inspired deep learning, agricultural automation, and clinical decision support. Skilled in AI automation workflows and agentic systems. Seeking an internship to apply research-driven problem-solving to real-world data challenges.

EDUCATION

Vellore Institute of Technology (VIT), Vellore *2023 – 2027 (Expected)*
B.Tech in Data Science & Engineering | **CGPA: 8.76 / 10.0**

- CFA Level I Candidate (CFA Institute, August 2026), demonstrating quantitative finance and analytical expertise.

Senior Secondary (Class XII) – CBSE *2023*
Percentage: 90.02%

Secondary (Class X) – CBSE *2021*
Percentage: 93.00%

PATENTS & INTELLECTUAL PROPERTY

Filed Patents (2 – Cleared VIT Stage 4 | Indian IPR Verification Ongoing):

- **Patent 1 –** A System and Method for Enforcing Privacy-Safe Execution of Database Queries with Cryptographically Verifiable Audit Control – Framework enforcing k-anonymity/l-diversity on SQL queries with Ed25519-signed cryptographic audit trails for healthcare environments.
- **Patent 2 –** Carbon-Aware Cloud Scheduling System with Adaptive Carbon Efficiency Index (ACEI) – Method for dynamically routing cloud workloads across global datacenters using real-time carbon intensity data with automatic failover and SLA compliance.

Upcoming Patents (4 in Progress, 2026):

- Hardware-Interfaced Spatio-Temporal Control Gating Apparatus and Adaptive Actuator Enforcement System for Agricultural Machinery – Cyber-physical control system integrating temporal gating with adaptive actuator enforcement for precision agricultural automation.
- Quantum-Inspired Adaptive Decision Routing (QIADR) – Deep learning architecture leveraging quantum-inspired superposition-based routing for dynamic neural network path selection.
- Resource-Adaptive Entropy-Gated Tiered Diagnostic Prediction Apparatus with Dynamic Model Routing and Explainable Feature Attribution for Clinical Decision Support – Multi-tiered diagnostic system with entropy-based gating for intelligent model selection and explainable AI in clinical workflows.
- AI-Driven Forensic Intelligence – Multi-modal analysis pipeline for automated cyber harassment investigation and digital evidence chain-of-custody verification.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C, R
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, spaCy, LangChain, Hugging Face Transformers, LLMs, YOLOv8, DeepSORT, RAG Systems
Frameworks & APIs: FastAPI, Next.js (React), Flask, Streamlit, Google Gemini API, Ollama
Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Recharts, Tableau, Power BI, Excel, Statistical Modeling
Databases: PostgreSQL, Supabase (pgvector), MySQL, MongoDB, SQLAlchemy
DevOps & Tools: Git, GitHub, Docker, Jupyter Notebook, Google Colab, VS Code, Linux/Bash
AI Automation: n8n, Make (Integromat), AI Agent Development, Workflow Orchestration, Prompt Engineering, Agentic Systems
Emerging Tech: Quantum Computing (Intermediate), Cloud Simulation (CloudSimPlus), Edge Computing

PROJECTS

ARGUS – AI-Powered Digital Forensics Platform *Hackathon Winner*
Next.js, FastAPI, spaCy, ResNet-18, NetworkX, Gemini 2.0 API, SHA-256 Hashing

- Engineered a multi-modal forensics pipeline ingesting social media posts, images, and PDFs to automate cyber harassment and identity theft investigations with a verifiable forensic chain of custody.
- Achieved 92.4% authorship attribution accuracy (Burrows' Delta) and 94.8% deepfake detection using a ResNet-GAN ensemble validated against Midjourney/DALL-E generated samples.
- Graph-based suspect identification via betweenness centrality (NetworkX), 85% confidence in isolating broker nodes; end-to-end processing under 1.5 seconds.

Repository: https://github.com/shlokDS16/ARGUS

C.I.T.A.D.E.L. – Civic Intelligence & Transparency Platform *Hackathon Project*
React (Vite), FastAPI, YOLOv8, DeepSORT, EasyOCR, Gemini Pro 1.5, Ollama/Llama 3.2, Supabase (pgvector)

- Architected a dual-portal (Citizen & Government) operating system integrating RAG-powered policy navigation, real-time traffic violation detection, and administrative automation.

- Hybrid RAG chatbot (Gemini Pro 1.5 + Llama 3.2 fallback) with 80%+ retrieval confidence across 500+ civic procedures; YOLOv8 + DeepSORT traffic module at 94.7% detection accuracy.
- EasyOCR document intelligence for ID extraction (under 1.5s), deep-learning fake news detection, and anomaly monitoring for civic data irregularities.

Repository: https://github.com/shlokDS16/CITADEL

PrivacyGuard Analytics – Privacy-Preserving Healthcare Platform *Academic Project*

Next.js 14, FastAPI, PostgreSQL, SQLAlchemy, Docker, Ed25519 Cryptography

- Full-stack platform enforcing k-anonymity and l-diversity on healthcare datasets; JIT + Heatmap strategy reduced privacy violations from 45 to 5 while retaining 64.2% data utility across 60 benchmark queries (UCI Heart Disease dataset).
- Ed25519-signed tamper-proof audit trail ensuring HIPAA-aligned regulatory compliance with sub-60ms P95 latency.

Repository: https://github.com/shlokDS16/Privacy-guard-

Carbon-Aware Cloud Scheduler – Sustainable Computing Optimizer *Research / Patent Project*

Java 17, CloudSimPlus, Apache Commons CSV, JFreeChart, OWID CO_2 Dataset (14M+ records)

- Patent-pending ACEI algorithm dynamically routing workloads across 10 global datacenters using real-time carbon data from 200+ countries; 15–25% carbon reduction, 8–12% cost improvement, 5–10% latency optimization with automatic failover.

CERTIFICATIONS & COURSES

- 100 Days of Code: The Complete Python Pro Bootcamp – Udemy (Dr. Angela Yu)
- The Complete Full-Stack Web Development Bootcamp – Udemy (Dr. Angela Yu)
- CFA Level I Candidate – CFA Institute (Exam: August 2026)
- Value Added Course: Quantum Computing – VIT Vellore (Ongoing)

ACHIEVEMENTS & AWARDS

- **Hackathon Winner –** Won the 36-Hour Cyber Forensics Hack-a-Thon for developing Project ARGUS, an end-to-end AI forensics platform.
- **Hackathon Winner –** Secured first place at AI Hackathon 2026: The Intelligent Innovation Challenge for building an AI-driven civic intelligence solution.
- **Academic Excellence –** CGPA of 8.76/10.0 across six semesters, consistently ranking in the top percentile of the Data Science cohort at VIT.
- **Early Research Recognition –** Two patents cleared through VIT Stage 4 review with Indian IPR verification in progress, plus four additional patents in preparation – a rare distinction for a 3rd-year undergraduate.

EXTRACURRICULARS & LEADERSHIP

IoT in Everyday Life Club, VIT Vellore *Board Member*

- Core board member contributing to strategy, event planning, and onboarding for a 100+ member technical community.
- Co-organized “Smart Campus Challenge,” an IoT hackathon attracting 80+ participants building sensor-driven prototypes.
- Led “IoT Meets AI” workshop series on integrating ML models with IoT devices using Raspberry Pi and edge computing.

Community Engagement

- Student mentor during VIT’s Freshman Orientation, guiding 30+ incoming students through academic planning and campus integration.
- Tech literacy drive teaching programming and digital skills to underprivileged high school students in Vellore district.